

Final PLL Proxy Report

Evidence source runs: full_run_20260617 and full_run_20260617_selected_handwritings.

Main claim: PLL can replace ground-truth-needed metrics for screening OCR outputs and choosing models when no transcription is available. It is a proxy for ranking, not a perfect substitute for a final WER/CER benchmark.

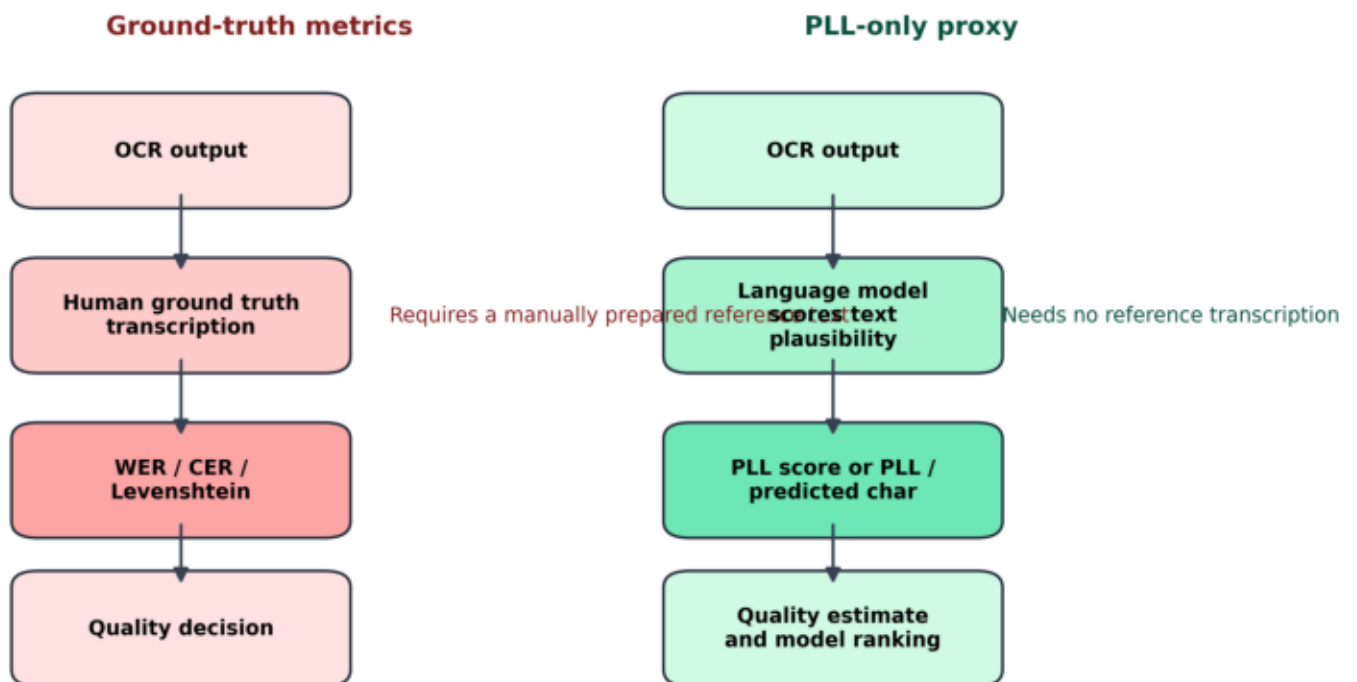
The operational no-GT scores are raw PLL and PLL per predicted character. Scores using GT length are kept only as validation checks.

Evidence	Comparison	Stat	Value	N
No-GT normalized page score	PLL / predicted char vs WER	Pearson r	0.794	224
No-GT absolute page score	Raw PLL vs Levenshtein	Pearson r	0.759	224
No-GT absolute validation ceiling	Raw PLL vs Levenshtein	Pearson r	0.922	351
Selected handwriting mean	Mean dataset PLL vs WER	Mean Pearson r	0.785	7
Selected handwriting mean	Mean dataset PLL vs CER	Mean Pearson r	0.759	7
Selected handwriting mean	Mean dataset PLL vs Levenshtein	Mean Pearson r	0.772	7
Selected model choice	PLL model rank vs WER model rank	Spearman rho	0.964	7
Full model choice	PLL model rank vs WER model rank	Spearman rho	0.857	7
Validation-only upper bound	PLL / mean GT-pred length vs WER	Pearson r	0.849	224
Full dataset equal-weight mean	Mean dataset PLL vs Levenshtein	Mean Pearson r	0.697	11

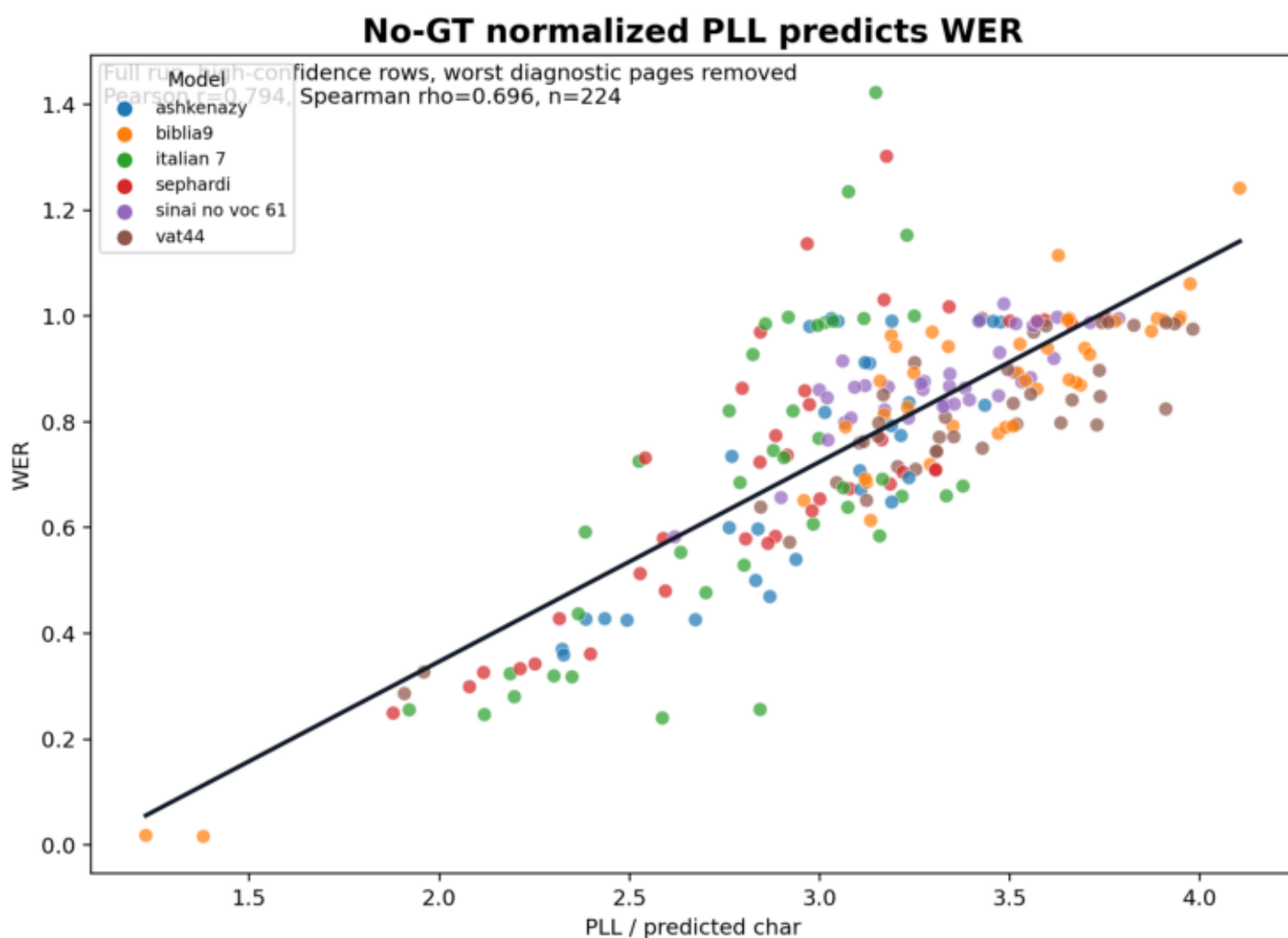
Scoring Workflow

GT metrics require reference transcription; PLL only requires OCR text and a language model.

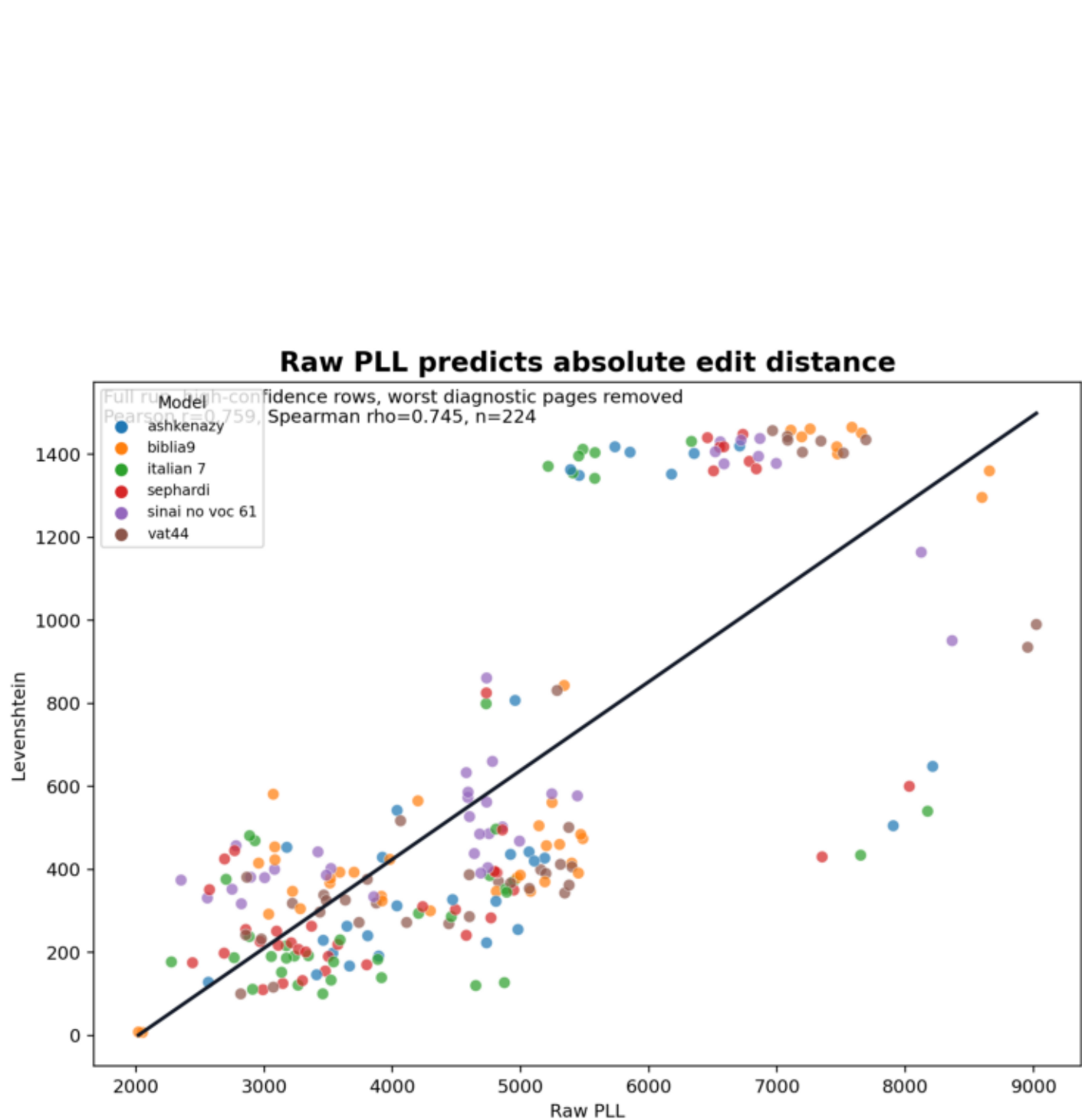
Two ways to score OCR quality



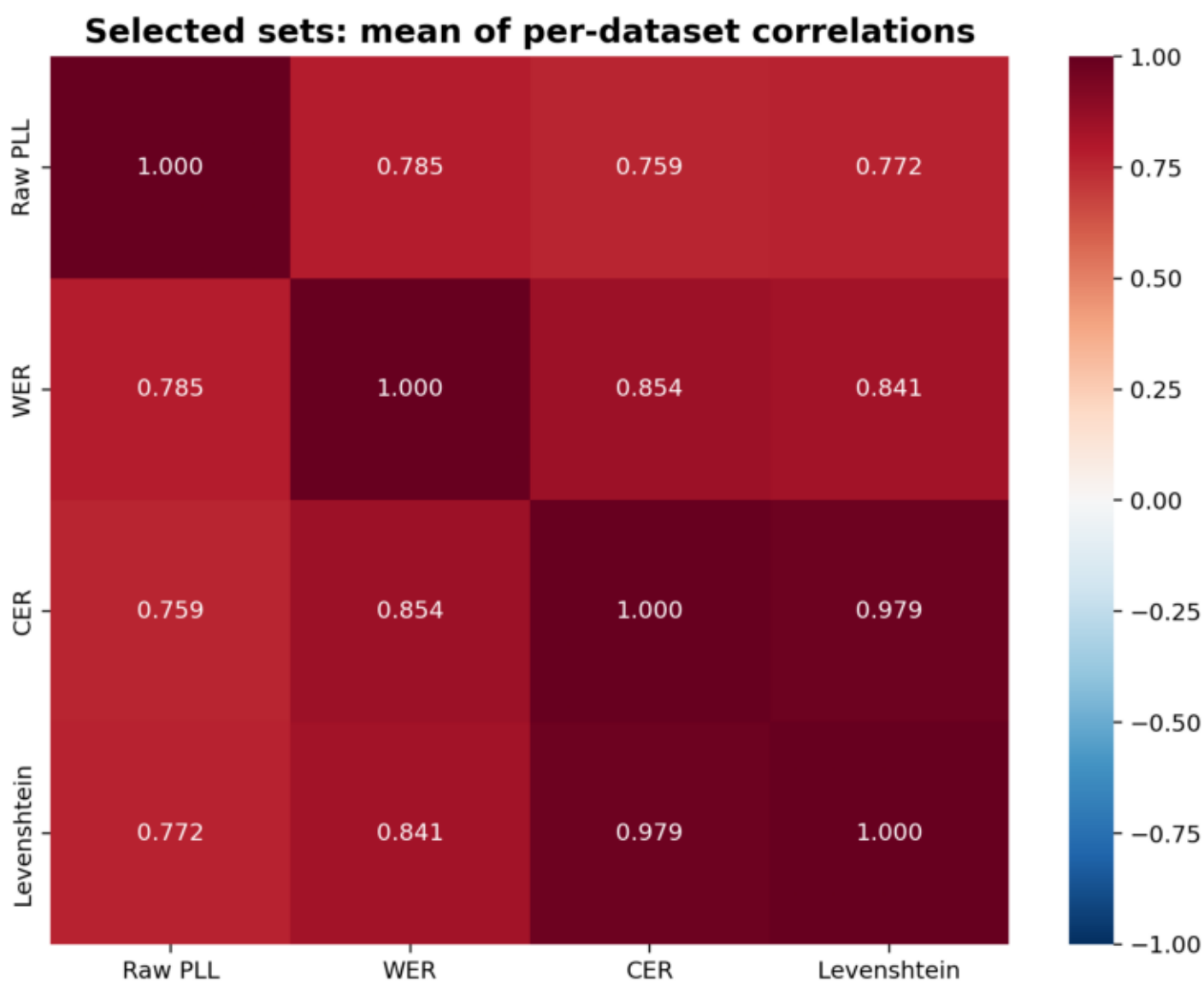
No-GT Normalized PLL vs WER



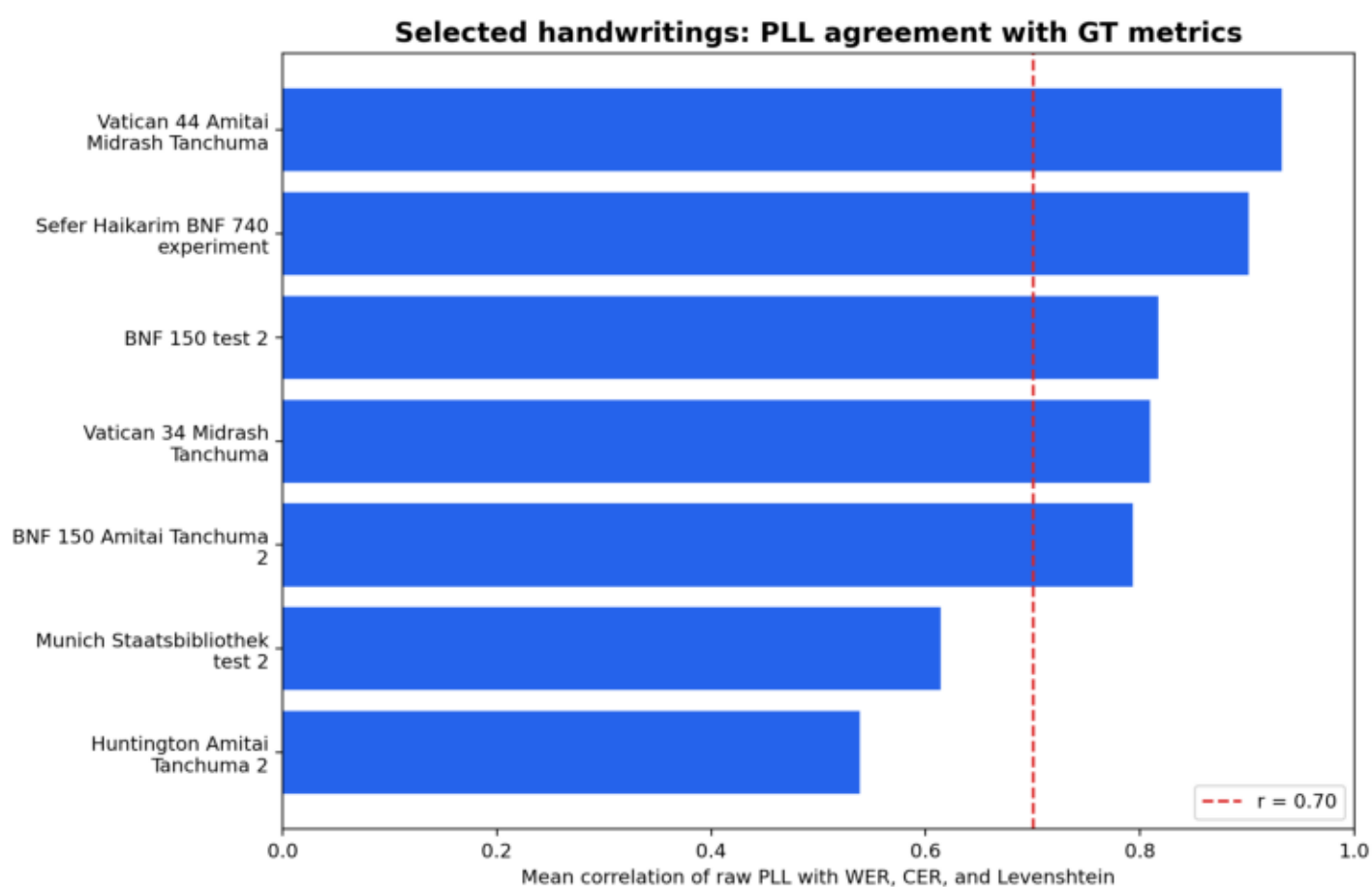
Raw PLL vs Levenshtein



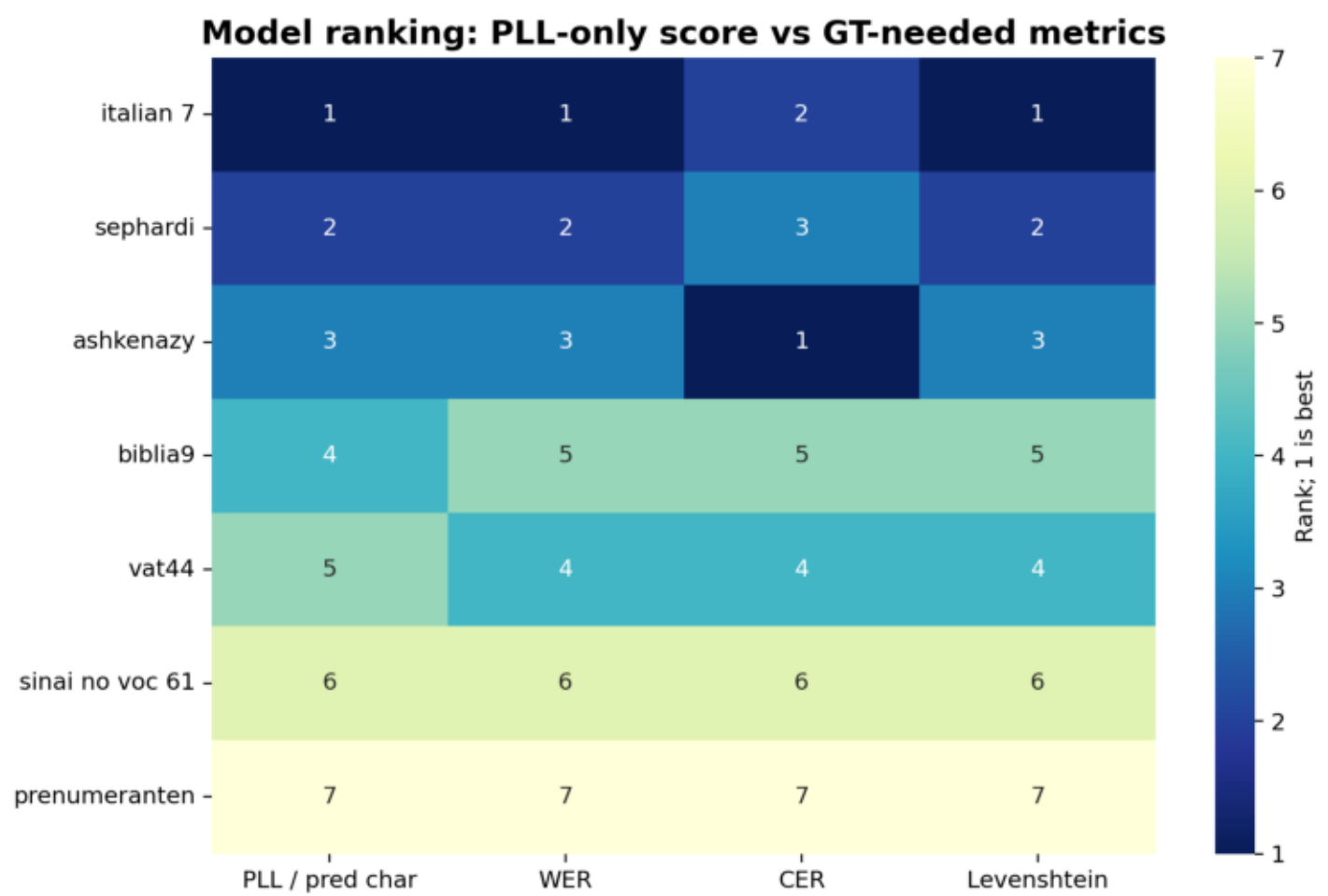
Selected Handwriting Mean Correlations



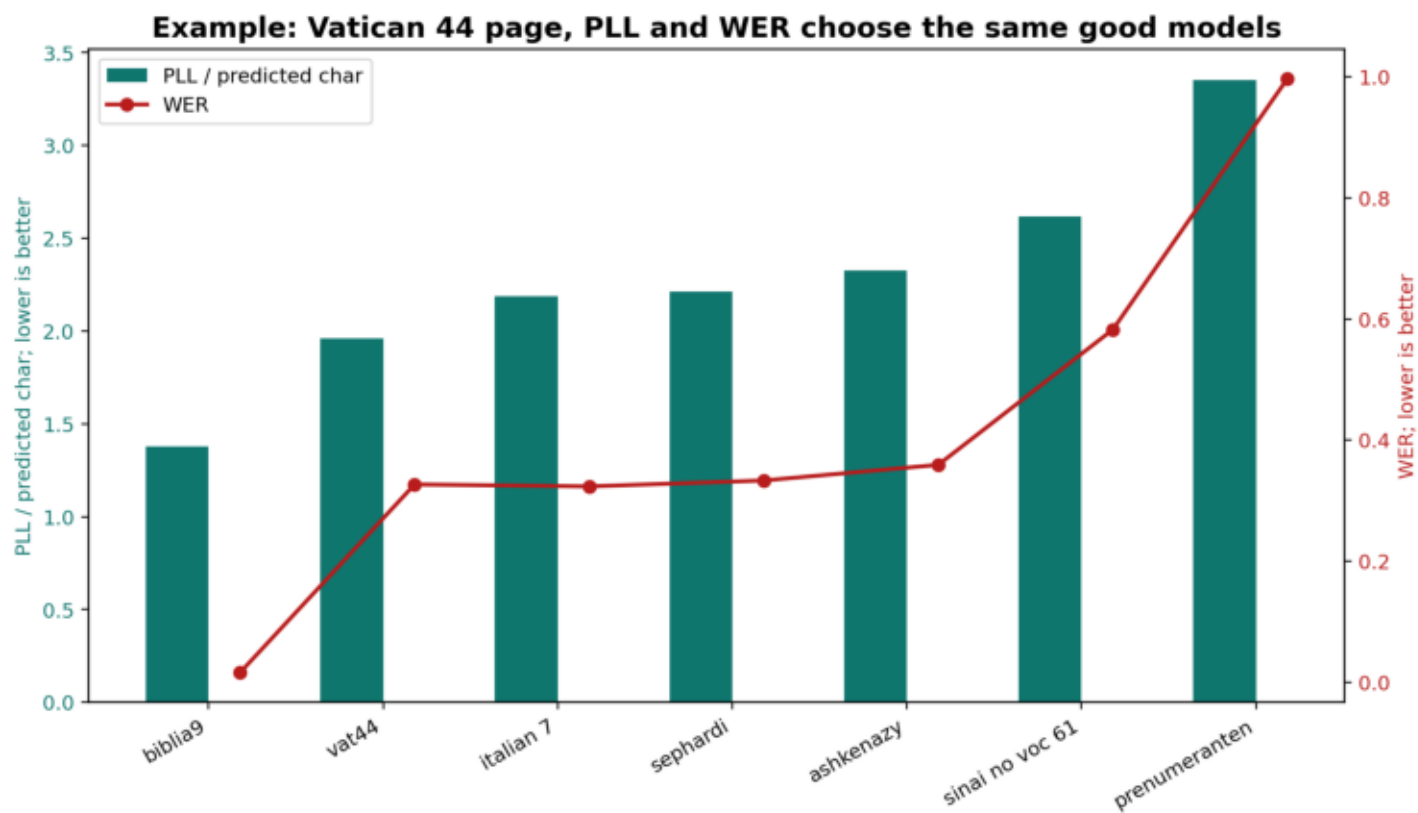
Per-Handwriting Agreement



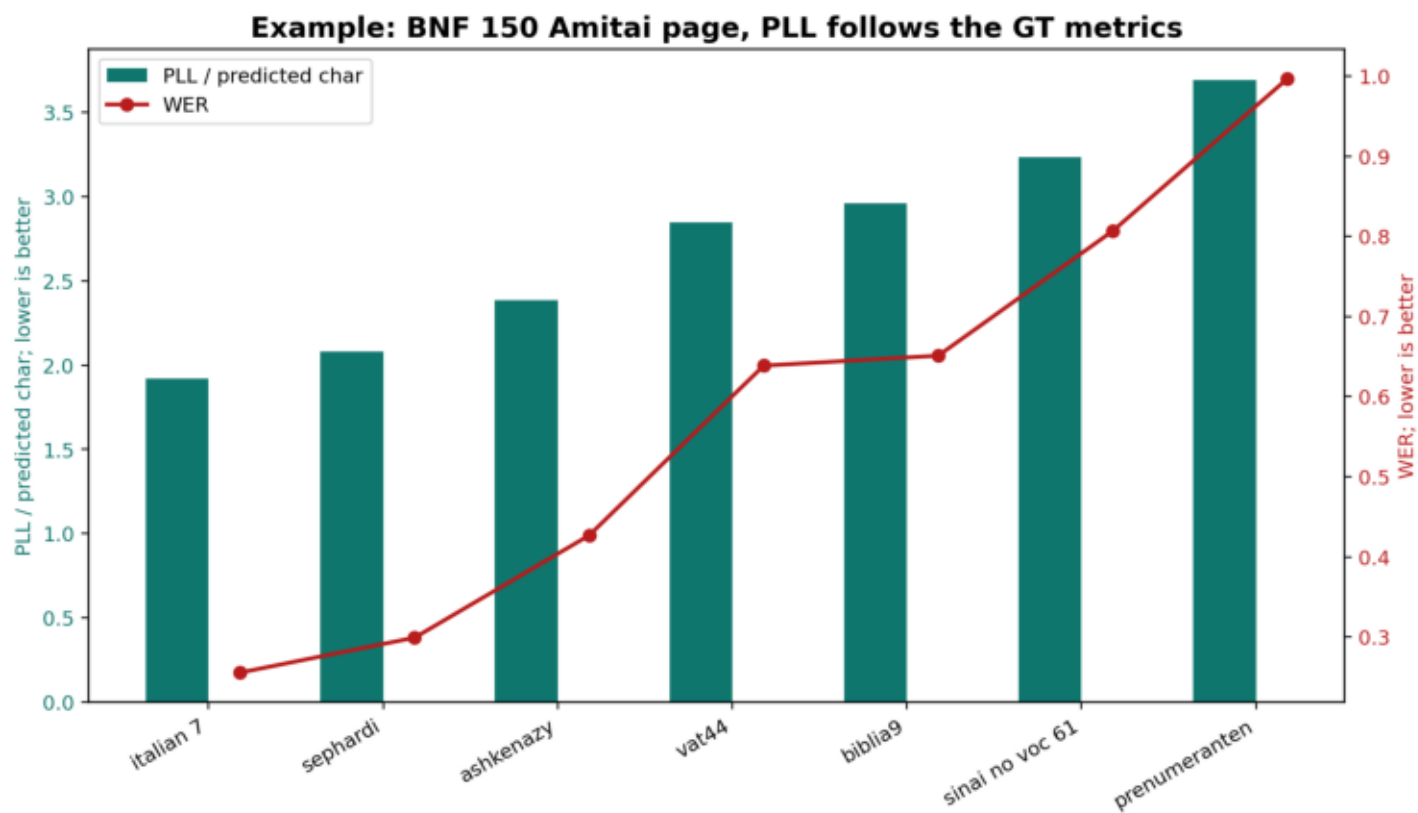
Model Ranking Agreement



Clean Example



Second Clean Example



Caveat Example

